UACE BIOLOGY PRACTICAL FACILITATION

PART 1:

NATURE OF THE PAPER AND THE ITEMS

1.1 Nature of the paper

- The duration of the paper is 3 hours and 15 minutes.
- The paper contains two compulsory items.
- The items are scenario based.
- Questions in this paper come from any of the four constructs addressing their respective assessment objectives.

1.2 The items

All items are scenario based; they are meant to improve critical thinking, creativity and problem solving.

ITEM 1: ***Observational scientific investigations i.e. comparative investigation item*****.** Examples are dissection and microscopy. This item addresses four competencies.

ITEM 2: ***Experimental scientific investigations i.e. scientific experimentation item.*** Examples are enzyme activity experiments, osmosis, etc. This item addresses five competencies.

1.3 Key to scoring levels used in this document

Code	**Meaning**	**Score**	**Remark**
AP	Highest level	3	Full competence shown
Ae	Moderate level	2	Competence partly shown
Al	Lowest level	1	Minimum competence shown

A tick (✓) in a model answer shows a scoring point. The letters T, M, A, L stand for Title, Magnification, Accuracy and Labelling in drawings; R, A, C stand for Relevance, Accuracy and Coherence in procedures.

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PART 2:

ITEM 1 – *OBSERVATIONAL SCIENTIFIC INVESTIGATION (comparative investigation item)*

2.1 What each candidate is expected to have

- Dissection set
- Pins
- A piece of thread
- Cotton wool
- Dissection board
- Clean paper and a sharp pencil
- Ruler
- Microscope
- Slides and cover slips
- Razor blade
- Stain, e.g. iodine

Required skills: observation, dissection, drawing and measurement.

2.2 The four assessment base codes of Item 1

- Rat dissection (DS).
- Biological drawings (DR).
- Dissection and microscopic measurements (DM).
- Suitability / adaptations (ST).

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Nature Of the Comparative Investigation Item

It involves collecting data from two animals.

- ✓ *One animal is hypothetical and data about it are also hypothetical. Support material may be used.*
- ✓ *The hypothetical animal has evolutionary disadvantages by virtue of data collected from its anatomy. The data about the hypothetical animal is provided in the scenario.*
- ✓ *The second animal, the rat, has superior characteristics for survival by virtue of data collected from its anatomy. This data is collected by the learner in the practical.*
- ✓ *The data collected is used to explain the differences in survival abilities of the two animals. (Suitability)*
- ✓ *The two animals may be related to a certain plant through any life process such as nutrition.*
- ✓ *This plant is to be examined under the microscope to obtain data that can explain its contribution to the survival of the animals.*
- ✓ *The plants include: monocotyledonous plants (roots, stems, leaves), dicotyledonous plants (roots, stems, leaves), Irish potato plant and onion plant.*

Bases Of Assessment for The Comparative Investigation Item.

- ➢ Measurements
- ➢ Procedure of microscopy
- ➢ Drawing from microscope
- ➢ Procedure of dissection
- ➢ Drawing from dissection
- ➢ Suitability for survival

SAMPLE ITEM 1A: RAT AND COMMON TOAD

A researcher at a Ugandan university investigated the factors influencing the survival of rats in grassland and newly metamorphosed common toads that dispersed from an adjacent wetland into the same grassland habitat over a four-month period. During the investigation, the researcher observed that the survival rate of common toads was drastically lower than that of the rats.

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The researcher hypothesized that the observed difference in survival rates of the two animals was partly associated with differences in the relative length of their digestive organs and dietary differences.

Field observations showed that both animals fed largely on invertebrates found in the grassland. However, the rats also fed on wild starch-rich tubers similar to specimen C whereas the adult common toads remained entirely carnivorous. The tubers similar to specimen C contain specialized food-storage structures that can be identified microscopically after staining with a suitable reagent. These observations indicate adaptations to survival.

To further investigate these adaptations, the researcher obtained data from a freshly killed common toad, recorded it in Table 1 and intended to collect corresponding data from a freshly killed rat (specimen E). However, the investigation was never completed.

Table 1: Length of digestive organs of a toad and a rat

Part of digestive system	***Toad***	***Rat***
Stomach (length in mm)	*48*	
Ileum (length in mm)	*150*	
Hind gut (length in mm), measured from end of small intestine	*58*	

The researcher also made drawings of the measured structures for a toad. To complete the investigation, there is need to compare data for the two animals together with drawings for the rat. You are therefore required to complete the investigation by obtaining data from specimen E together with making drawings. You are provided with a freshly killed rat labelled E.

Tasks.

(a) *Obtain data from specimen E and complete Table 1. (Your responses should include a procedure for carrying out the dissection.) Make drawings of the measured structures.*

(b) *Provide microscopically obtained evidence to support the nutritional advantage of this food source the rats have over toads in the grassland.*

(c) *Analyze the findings from this whole investigation (your data and the researcher's data) to justify the differences in the survival abilities of the common toad and specimen E (rat).*

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RAT DISSECTION (DS)

Required competencies in rat dissection

1. Dissection procedure:

a) **Orientation:** laying the specimen and pinning.

b) **Opening up the animal:** slitting the skin, the mid-ventral line cut, the abdominal wall cut and pinning.

c) **Exposing the relevant structures (**Displacements to display relevant structures**)** It is vital to know which structures are relevant to the scenario.

General dissection procedure

(a) Opening the abdomen

- The rat was placed with the ventral side uppermost on the dissection board and pinned through the feet.
- Skin in the mid-ventral line was lifted and an incision made; the cutting was continued forward to the lower lip and posteriorly up to the genitals.
- Skin was loosened and pinned to expose the body wall.
- An incision was made along the mid-ventral line of the abdominal body wall and the cutting was continued up to the xiphoid cartilage, the genitals and along the diaphragm.
- The body wall was pinned so as to expose the abdominal viscera.

(b) Opening up the thorax and neck

- Grip the xiphoid cartilage and cut through the posterior part of the thoracic wall.
- Cut through the side wall of the thorax.
- Continue with the cutting to the apex of the thorax, turning the ventral part of the wall aside.
- Repeat on the other side to remove the ventral part of the thoracic wall entirely.
- Cut away loose parts.
- Be sure to cut through the anterior ribs and to remove the sternum completely.

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(c) Removal of the alimentary canal

The alimentary canal must be removed before any dissection of the urinogenital system and abdominal blood vessels.

- Ligate the coeliac and anterior mesenteric arteries and the hepatic portal vein.
- Cut distal to the ligature.
- Cut the esophagus.
- Cut the rectum.
- Gently tear the mesentery to remove the alimentary canal.

(d) Urinogenital system

- The alimentary canal must be removed first.
- Remove fat from the kidneys and ureters.
- Cut through the ventral wall of the scrotal sac.
- Open the scrotal sac.
- Cut through the pelvic girdle.
- Displace the bladder to one side.
- Displace the seminal vesicles, coagulating gland and prostate glands to one side.

(e) Heart and great blood vessels

- Pin the heart to the right.
- Remove the thymus gland.
- Clear away fat from around the blood vessels.

NOTE

- *The major regions to be examined are: the abdominal viscera and Thoracic viscera.*
- *Only the major blood vessels to the particular organs are to be measured.*
- *To expose the stomach, the liver is displaced anteriorly.*
- *To expose the ileum, duodenum is pulled to the right of the specimen (i.e. your left)*
- *Always include what you are cutting and what you are exposing.*
- *Hepatic portal vein is measured from the liver to any nearby organ.*
- *Carotid artery is measured from the heart to the head (masseter muscles)*

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Marking guide for dissection (DS)

Marking points (each tick ✓ is a point):

- *Specimen was placed with the ventral side uppermost on the dissection board and pinned through the feet;* ✓
- *Skin in the mid-ventral line was lifted and an incision made and the cutting was continued forward to the lower lip and posteriorly up to the genitals;* ✓
- *Skin was loosened and pinned to expose the body wall;* ✓
- *An incision was made along the mid-ventral line of the abdominal/body wall and the cutting was continued up to the xiphoid cartilage, genitals and along the diaphragm;* ✓
- *Body wall was pinned so as to expose the abdominal viscera;* ✓
- *Duodenal loop was pulled to the right, bulk of the ileum to the left, and caecum and colon downwards and to the right;* ✓
- *Liver lobes were anteriorly displaced to clearly display the stomach;* ✓

Competence	***Scoring***
Body orientation	*Ae: ventral orientation and pinning. Al: ventral or pinning.*
Dissection	*AP: 4 displacements/displays of liver, ileum, hind gut and viscera mentioned. Ae: 2–3 displacements/displays. Al: 1 displacement.*
DS total	*1 AP = 3; Ae = 2; Al = 1*

BIOLOGICAL DRAWINGS (DR)

The competencies assessed in a drawing are:

1. Title (T). i.e. *A drawing of a (specific part talked about in the scenario) of specimen….*
2. Magnification (M). i.e. x1-3 in a rat.
3. Accuracy (A): continuous outline, proportionality and neatness.
4. Labelling (L). i.e. be theory based.

***Note**: There should be at least two drawings, one from dissection and the other from microscopy. All drawings are assessed under the same base code.*

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Drawing from dissection: parts of the alimentary canal of the rat

The drawing should show the stomach, ileum, colon and rectum, correctly labelled. Marks are given for title (T – 1), magnification (M – 1), accuracy (A – 1: outline, proportionality, neatness of drawing) and labelling (L).

Drawing from microscopy: one cell of specimen C showing stored food

Title: Drawing of one cell of specimen C showing stored food as observed under low or medium power of a microscope.

The drawing shows a cell with its two layers close to each other with only a small gap, and the starch grains/granules labelled. T – 1, M – 1, A – 1, L – 1.

Competence	***Scoring***
Attributes indicated	*AP: 5–8 attributes. Ae: 3–4 attributes. Al: 1–2 attributes.*
DR total	*AP = 3; Ae = 2; Al = 1*

Note:

- ✓ *The two cell layers are close to each other with only a small gap.*
- ✓ *Procedure of microscopy: The learner is expected to write the complete and detailed procedure from slide preparation to microscope use.*

DISSECTION AND MICROSCOPIC MEASUREMENTS (DM)

- With dissection measurements, candidates are encouraged to use a piece of thread and a ruler, or a ruler only where possible.
- The units must be considered: any candidate who uses units different from those in the scenario is not scored as correct.
- The measurements must be in the range; those out of range are not scored.
- Candidates are encouraged to read and follow what is required so as not to miss out on these scores.

Note: *The competency here is the ability to measure and record correctly*.

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Dissection measurements of the rat (Table 1):

Part of alimentary canal	***Length (mm), acceptable range***
Stomach	*60 – 80*
Ileum	*500 – 1200*
Hind gut	*140 – 190*

Scoring: AP – 3 correct measurements; Ae – 2 correct measurements; Al – 1 correct measurement.
DM: AP = 3; Ae = 2; Al = 1.

Measurements

These can be obtained from:

- ✓ ***The external features of the rat such as tail length, body length, eyes, ears, head circumferences, vibrissae …***
- ✓ ***Internal structures such as length of ileum, stomach, duodenum, caecum, heart, lungs, blood vessels …***

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SUITABILITY / ADAPTATIONS (ST)

- Suitability of the specimen depends on the measurements taken and the information given in the scenario.
- Relate the measurements to solving the problem in the scenario, adaptation and survival.
- Candidates who give wrong measurements often lose out in this base code.

NOTE: *ST focuses on explaining the differences in survival abilities of the two animals in their environment based on data collected; this explanation must be clearly linked to the scenario.*

(c): Analyze the findings to justify the differences in the survival abilities of the common toad and specimen E (rat).

- *The rat has a longer ileum/stomach; to increase surface area and allow longer time for digestion/absorption of nutrients from digested food;* ✓
- *The longer hind gut of the rat increases surface area for absorption of much water from undigested food; thus, the rat is better adapted for water conservation;* ✓
- *The rat has both sole pads and claws, which greatly increase grip, compared with the toad with only sole pads. This grip allows faster locomotion and therefore faster escape speed from predators and faster ability to find food, increasing the survival ability of the rat;* ✓
- *The ability of the rat to feed on both animal protein and starch from specimen C increases its chances of finding food, unlike toads which depend on one source of food;* ✓

Competence	***Scoring***
Explanation for survival	*AP: explanation includes adaptations on at least 3 relevant structures (ileum/stomach, hind gut, sole pads/claws, starch from specimen C). Ae: explanation includes adaptations on at least 2 relevant structures. Al: explanation includes adaptation on any one.*
ST total	*AP = 3; Ae = 2; Al = 1*

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SAMPLE ITEM 1B: SPECIMENS K, L, M AND N (KADAMA VILLAGE)

In Kadama village, farmers practice mixed farming of specimen K (a root crop) and specimen L (a leafy plant). They have reported to an agricultural officer a decline in the yields of specimen K due to selective pest damage, especially during harvesting, while specimen L remains largely unaffected. Field observations suggest that pests preferentially feed on specimen K.

To investigate this, the officer collected specimens K and L for microscopic study using iodine. In addition, two pest variants were identified in the farmers' gardens: specimen M and N. Specimen M eats only a part of the tubers of K but does not finish, does not easily detect when being approached, gets stuck halfway in the holes and also runs slower than specimen N. These make specimen M easily caught and killed. The officer also studied specimen M and obtained the following results.

Parameter	***Results of specimen M***	***Results of specimen N***
Length of the longest whiskers	*2 cm*	
Circumference of the hind limb thigh	*0.5 cm*	
Dental formula	*I 1/1, C 0/0, P 0/0, M 3/3*	
Length of the right subclavian artery up to elbow	*10 cm*	
Distance of the eye from the dorsal surface	*10 mm*	
Length of innominate artery	*3 cm*	
Length of the broadest part of the pinna	*10 mm*	
Length of ventricle	*4 mm*	

The officer concluded that differences in plant tissue structure and nutrient storage influence pest feeding, and that the differences between specimens M and N explain their differences in behavior, particularly in feeding efficiency, movement and ability to avoid capture. However, the officer was unable to obtain specimen N at the time.

He therefore requires comparative findings on specimen N and a drawing of its blood supply to the head region and limbs, and a drawing of the tissue of specimen K for record keeping.

You are provided with specimens K, L and N.

Note: *The task questions of this sample item are not visible in the scanned slides.*

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PART 3

ITEM 2 – EXPERIMENTAL SCIENTIFIC INVESTIGATION

3.1 Introduction to Item 2

- A foundation was already created in the O-level practical.
- Item 2 has five assessment base codes, each with its competency.
- Candidates are expected to have a theory background of the investigations.

New additions:

- The aim must have the what and the why, with a reason for the experiment.
- Two independent variables are tested.
- The hypothesis must have predictions with reasons.
- Variables must be categorized.
- Risks and mitigations are required.

Bases of assessment for the scientific experimentation item

- Planning
- Risks, ~~sources of errors~~ and their mitigation
- Procedure of experiment
- Data presentation
- Data analysis
- Recommendation and advice

3.2 The five assessment base codes of Item 2

No.	Base code	Abbreviation
1	Planning experiment (aim, hypothesis, variables)	PE
2	Method of experiment (risks and mitigations, procedure)	ME
3	Data presentation	DP
4	Data analysis	AN
5	Conclusion and recommendation	CR

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PLANNING EXPERIMENT (PE)

(a) Aim of the experiment

- The aim must address the ***what*** and the ***why***. Learners are expected to have background theory knowledge.
- What: states the independent and dependent variables (for both factors being investigated).
- Why (Purpose for experiment): states the reason for the experiment (it is obtained from the scenario).
- Highest level: the what and the why are stated with reason.

Example: *An experiment to investigate the effect of concentration of solution enzyme from ileum extract and pH of substrate in solution D; so as to determine which ratio of water to enzyme extract and pH solution that can produce the best results.*

(b) Hypothesis

A hypothesis is a prediction/relationship between the independent and dependent variables. It states what you expect the independent variable to do to the dependent variable, and the reason for the expectation. It must have two predictions with reasons. It has three components:

- What did you do? (independent variable)
- What are you expecting to see? (dependent variable)
- Why are you expecting what you expected? (reasons)

Example: *The ratio of enzyme extract to water of 5:1, and pH solution A, produce better results than ratios 4:2, 3:3, 2:4 and 1:5, and pH solutions B and C causing production of more sugars because at that concentration the enzyme and substrate molecules are closer to each other and at that pH the enzyme is most activated.*

(c) Variables

- **Independent variables:** What learners are manipulating to cause variation in results (the cause of the results; in this case the solutions provided are the causes of the results.) E.g. ratios of water to enzyme extract / concentration of enzyme
- **Dependent variables:** Outcome as seen in the scenario (the expected results of the experiment). E.g. amount of sugars formed / colour changes of solution.
- **Controlled variables:** (what is kept constant in the experiment in order to have consistent results). Volumes of solutions A, B and C; time for incubation, temperature of incubation

NOTE*: Scoring: highest level – all three categorized; moderate level – two categorized; lowest level – one category corrects; no score – no/wrong categories.*

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METHOD OF EXPERIMENT (ME)

(a) Risks and mitigations

- Candidates are required to identify the possible risks during the experiment and give measures on how to avoid them.
- Identify possible risks basing on what has been provided for that particular experiment, e.g. a knife, razor blade, test tube, acid or peroxide.
- A knife, razor or broken test tube can lead to cuts as a result of careless handling. It is not advisable to identify each one independently.
- Peroxide can lead to skin irritation because of corrosion.
- ***Risks and mitigation: example****: Burns from hot test tube and hot water mitigated by Holding test tube with test tube holder.*

NOTE: *Scoring: Ae – 2 risks identified and mitigated; Al – 1 risk identified and mitigated, or risks identified with no mitigation.*

(b) Procedure: it must meet R, A and C

- **R – Relevant:** the procedure given must be able to solve the two problems in the scenario. Each procedure indicated is assessed independently of the next one and must lead to results. It is advisable to write the procedure in paragraphs to avoid confusion when performing the experiments. It is vital to label test tubes, beakers, etc. when carrying out experiments.

  Scoring*: highest level – solves two problems; moderate level – solves completely one problem; lowest level – partially solves one problem; no score – procedure cannot lead to solving any problem*.

- **A – Accuracy of measurements:** candidates are advised to take note of the measurements during the experiment. Measurements here include volumes, length, etc., depending on the experiment. Ratios are key when dealing with measurements.

  Scoring: *highest level – uniform measurements of similar materials/ratios of solutions used; moderate level – one wrong ratio/measurement; lowest level – measurements given but no ratios; no score – no measurements given.*

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- **C – Coherence:** the sequence of addition of solutions. Candidates are required to know which solution to add first then next. This determines the results of the experiment. Candidates who do not provide the correct sequence of addition of solutions cannot obtain correct results. (For instance; under enzyme activity: substrate+regulator(factor)+enzyme)

  Scoring: *highest level – all correct sequence; moderate level – one wrong sequence; lowest level – all required solutions mixed; no score – not all required solutions mixed.*

NOTE:

- *Some steps of the procedure are guided by the scenario.*
- *Measurements such as time, temperature, volumes and length are dictated by the scenario to avoid extreme variations in results.*

DATA PRESENTATION (DP)

- Data must be correct and the format must be user friendly and easy to identify trends.
- Data should be presented in a table.
- This base code includes experimental observations.
- A candidate must have a correct and coherent procedure to score all observations for both experiments.
- Data can also be further analyzed using a graph for better meaning i.e. if the graph is in the question paper.

Scoring: *Level 3 – results solve two problems; Level 2 – results solve one problem; Level 1 – partial results towards solving a problem; no score – results cannot solve any problem.*

DATA ANALYSIS (AN)

- Must have two trends/factors, explained biologically.
- This base code requires a detailed explanation of the processes under investigation.
- Candidates are expected to give biological explanations of what happened during the investigation.
- Data analysis requires full theory knowledge.

Scoring*: Level 3 – two trends given and explained; Level 2 – one trend given and explained; Level 1 – two or one trend given and not explained; no score – no trend given.*

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NOTE:

Involves a deeper explanation of physiological processes:

1. *The physiology of enzymes, lock and key hypothesis, active site, competitive and non-competitive inhibition, denaturation of enzyme by temperature and Ph (i.e. the specificity of enzymes). i.e. (enzymes from ileum and stomach)*
2. *Physiology of events leading to germination.*
3. *Physiology* of osmosis.

The explanations must finally be linked back to the scenario.

CONCLUSION AND RECOMMENDATION (CR)

- Must have two conclusions from the two independent variables under investigation and an appropriate recommendation addressing the problem in the scenario.
- Conclusion: give the conclusion while addressing the stated hypothesis.
- Recommendation: let the recommendation address the identified problem in the scenario.
- Give reason for the recommendation

Scoring*: Level 3 – two conclusions made and a recommendation made; Level 2 – one conclusion made and a recommendation made; Level 1 – two or one conclusion(s) made; no score – no conclusion and no recommendation made.*

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SAMPLE ITEM 2A – JOAN'S GAS PRODUCTION

Joan is piloting a project in which she intends to generate and sell gas to medical facilities for usage by patients in intensive care units. She produces the gas by the catalytic breakdown of solution P under constant temperature. She was advised to use an organic catalyst/enzyme which is obtained from plant extracts.

Joan obtains her enzyme from extract C, which is obtained by mixing 4 g of peeled and crushed Irish potato with 50 cm^3 of distilled water; the mixture is stirred and decanted.

However, she is not sure whether the extracts on the market labelled A and B could produce more gas per unit time than her local extract C.

To further increase gas production, Joan was advised to add an enzyme activator. Currently there are two types of enzyme activators on the market, solution X and Y, and she needs to be guided on which one increases gas production.

You are provided with solution P, enzyme activators X and Y, enzyme extracts A and B, then specimen C from which Joan prepared her enzyme extract C. Joan needs guidance on how to select the appropriate materials for her gas production.

Task:

Design and carry out a scientific investigation on gas production from P using the available materials and use your findings to give appropriate advice to Joan.

What is required in the task: give the aim, hypothesis, variables, risks and mitigations, procedure, data presentation, data analysis, and conclusion and recommendation. (Your report should include explanations.)

Planning experiment (PE)

Aim: Experiment to determine which of enzyme extract A or B ✓ and activators X or Y ✓ produce more gas per unit time on reacting with substrate ✓ in solution P so as to recommend to Joan ✓.

Scoring*: AP – 2 experiments on enzyme extract concentration and activator with reason and substrate mentioned (4 ticks); Ae – 1 experiment with reason, or 2 experiments with no reason but with substrate mentioned (3 ticks); Al – 1 experiment with no reason but with substrate mentioned (2 ticks).*

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Hypothesis: Enzyme extract A and activator X together with substrate in solution P produce more gas because extract A contains the highest enzyme concentration and activator X promotes faster formation of enzyme-substrate complexes.

Components: independent variable – enzyme extract A and activator X; dependent variable – produce more gas on reacting with substrate in solution P; reasons – extract A contains the highest enzyme concentration and activator X promotes faster formation of enzyme-substrate complexes.

***Scoring**: AP – 2 predictions with 2 reasons; Ae – 1 prediction with reason, or 2 predictions with no reason; Al – 1 prediction with no reason.*

Variables:

Category	**Variables**
Independent	Enzyme concentrations in solutions A, B and C; activators X and Y
Dependent	Rate of gas production / rate of effervescence / bubbling
Controlled	Volume of enzyme extracts A, B and C; volume of activator solutions X and Y; volume of substrate in solution P; temperature; duration of the experiment

***Scoring:** AP – 5 or more variables categorized in 3 groups; Ae – 3–4 variables categorized in 2 groups; Al – 1–2 variable(s) categorized in 1 group. PE: 2/3 AP = 4; 1 AP = 3; Ae = 2; Al = 1.*

Method of experiment (ME)

Risks and mitigations

- Cuts from careless handling of a knife/sharp instruments; mitigated by careful handling of sharp instruments. ✓
- Skin irritations when corrosive reagents get into contact with the skin; mitigated by wearing protective gloves/lab coats. ✓

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Procedure 1:

Effect of enzyme concentration

- Three test tubes were labelled A, B and C; ✓
- To each test tube, 2 cm^3 of solution P were added; ✓
- To test tube A, 2 cm^3 of extract A were added and observations recorded; ✓
- The previous step was repeated for each of test tubes B and C with their corresponding enzyme extracts. ✓

Procedure 2:

Effect of activators

- Two test tubes were labelled X and Y; ✓R
- To each test tube, 2 cm^3 ✓A of enzyme extract A were added; ✓R
- 1 cm^3 ✓A of the respective activator was added to the corresponding test tubes; ✓R
- 2 cm^3 ✓A of solution P were added to each test tube ✓R and observations were recorded. ✓C

Scoring: *R – AP: 5–7 relevant steps; Ae: 2–4; Al: 1. A – AP: 3–5 accurate measurements; Ae: 2; Al: 1. C – Ae: 2 coherent steps; Al: 1 coherent step. ME: 2 AP = 4; 1 AP = 3; Ae = 2; Al = 1.*

Data presentation (DP)

Test tube	Observation
A	Rapid effervescence
B	Moderate effervescence
C	Slow effervescence
X	Rapid/high effervescence
Y	Low or no effervescence

Scoring: *AP – 5 correct observations; Ae – 2–4 correct observations; Al – 1 correct observation. DP: AP = 3; Ae = 2; Al = 1.*

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Data analysis (AN)

- As enzyme concentration increases, the frequency of collisions between enzyme and substrate molecules increases, resulting in an increase in the rate of formation of enzyme-substrate complexes, thereby causing an increase in the rate of breakdown of substrate; hence an increase in the rate of effervescence.
- Activator solution X modified the enzyme/substrate shape for faster formation of enzyme-substrate complexes, resulting in faster breakdown of substrate; hence a higher rate of effervescence.
- Solution Y provided an unfavourable medium, thereby inhibiting enzyme activity.

Scoring: *AP – 5 or more explanation points; Ae – 3–4 explanation points; Al – 1–2 explanation points. AN: AP = 3; Ae = 2; Al = 1.*

Conclusion and recommendation (CR)

- **Conclusion:** Extract A ✓ and activator X ✓ produced the highest rate of gas production on reacting with substrate, as predicted.
- **Recommendation:** Joan should combine enzyme extract A ✓ and activator X ✓ and react with substrate in solution P, as this combination produces the highest rate of effervescence.

Scoring: *AP – the appropriate enzyme and activator mentioned in both conclusion and recommendation; Ae – the appropriate enzyme and activator mentioned in the conclusion or the recommendation; Al – the appropriate enzyme or activator mentioned in the conclusion or the recommendation. CR: AP = 3; Ae = 2; Al = 1.*

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SAMPLE ITEM 2B – VILLAGE HEALTH TEAM AND WASTE BREAKDOWN

A village health team was inspecting one of the local markets and found wastes accumulating in the only water source in the area, which could cause a disease outbreak at any time to the local people. The team leader came up with an idea of using aerobic organisms to break down the wastes. However, upon introducing the organisms, the team realised that the rate of breakdown of the wastes was slow.

When they consulted the district environmental officer, he advised them to use an active substance and hydrogen peroxide solution, which yield oxygen when reacted together. This oxygen can boost the activity of the aerobic organisms. He informed them that the higher the concentration of the active substance and hydrogen peroxide solution, the more the oxygen produced and hence the higher the activity of the organisms.

When the team went to the shop to obtain the required active substance and the peroxide, they found three brands of the active substance labelled E, F and G and two brands of hydrogen peroxide labelled J and K. The team consulted the …

Note: The investigation uses pieces of filter paper soaked in solutions E, F and G, which rise to the surface of solutions J and K when oxygen bubbles are produced.

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Planning experiment (PE)

Aim: Experiment to determine which of the solutions of active substance E, F and G ✓ and substrate solutions J and K ✓ on reacting produce more oxygen, leading to a shorter time of the filter paper rising to the surface, so as to recommend to the health team. ✓

Hypothesis: Solution of active substance G ✓ and solution of substrate J ✓ on reacting produce more oxygen, causing the filter paper piece to rise faster, because solution G has the highest concentration of the active substance ✓ and solution J has the highest concentration of substrate. ✓

Variables:

Category	**Variables**
Independent	Concentration of active substance in E, F and G; substrate concentration in J and K
Dependent	Rate of movement of the piece of filter paper; rate of bubbling
Controlled	Volume of active substance solutions E, F and G; volume of substrate solutions J and K; size of filter paper used; time taken for the filter paper piece to rise (as listed on the slide)

Method of experiment (ME)

Risks and mitigations

- Cuts from careless handling of a knife/sharp instruments; mitigated by careful handling of sharp instruments. ✓
- Skin irritations when corrosive reagents get into contact with the skin; mitigated by wearing protective gloves/lab coats. ✓

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Procedure

- Label 3 petri dishes as E, F and G; ✓R label 6 test tubes as 1, 2, 3, 4, 5 and 6; ✓R
- Add 5 cm^3 ✓A of solutions E, F and G to each respective petri dish; ✓R
- Dip 2 pieces ✓A of filter paper into each solution in the petri dishes ✓R and leave them for 5 minutes; ✓A
- Add 5 cm^3 ✓A of solution J to test tubes 1, 2 and 3; ✓R add 5 cm^3 of solution K to test tubes 4, 5 and 6; ✓R
- Gently dip one piece of filter paper from petri dish E into test tube 1 and record the time taken to rise to the surface; ✓R ✓C
- Repeat the previous step using filter paper from E to 4, F to 2 and 5, and G to 3 and 6.

Data presentation (DP)

Time taken for the filter paper to rise to the surface:

Test tube	Time taken
1	20
2	15
3	6
4	28
5	17
6	10

Data analysis (AN)

The data analysis slides in the scan are for Joan's investigation (see section 4.6); the same enzyme-concentration principle applies here: a higher concentration of active substance and substrate increases collisions, forms more complexes, and produces oxygen faster, so the filter paper rises in a shorter time (test tube 3, solution G with J, rose in 6 units of time).

Conclusion and recommendation (CR)

Conclusion: Active substance solution G ✓ and substrate solution J ✓ produced more oxygen on reacting, causing a faster rise of the piece of filter paper, as predicted.

Recommendation: The health team should use solution G ✓ and substrate solution J ✓ since they produced more oxygen on reacting, causing a faster rise of the piece of filter paper.

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SUMMARY OF SCORING FOR ALL BASE CODES

Item	Base code	Scoring (AP / Ae / Al)
Item 1	DS – Rat dissection	3 / 2 / 1
Item 1	DR – Biological drawings (T, M, A, L)	3 / 2 / 1
Item 1	DM – Dissection and microscopic measurements	3 / 2 / 1
Item 1	ST – Suitability / adaptations	3 / 2 / 1
Item 2	PE – Planning experiment	AP = 3 or 4; Ae = 2; Al = 1
Item 2	ME – Method of experiment	AP = 3 or 4; Ae = 2; Al = 1
Item 2	DP – Data presentation	3 / 2 / 1
Item 2	AN – Data analysis	3 / 2 / 1
Item 2	CR – Conclusion and recommendation	3 / 2 / 1

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